

SYSTEM ANALYSIS AND DESIGN

Professional Portfolio Website

SAD Assignment Report

Student Name	K.M.N.M.R.S.R. Thilakarathna
Registration Number	2022/ICTS/45
Degree	Bachelor of Information and Communication Technology (Hons)
University	University of Vavuniya
Live Website	https://shenalthilakarathana2003-eng.github.io/SAD-Portfolio/
GitHub Repository	https://github.com/shenalthilakarathana2003-eng/SAD-Portfolio
LinkedIn	https://www.linkedin.com/in/shenalthilakarathne-9539a7343

Executive Summary

This report presents the analysis, design, implementation and evaluation of my professional portfolio website. I developed this website as a Bachelor of Information and Communication Technology (Hons) undergraduate at the University of Vavuniya. The main purpose of the project is to provide one professional online space where my academic background, technical skills, projects, experience, CV and contact information can be presented clearly.

The portfolio is implemented as a responsive single-page front-end website. HTML5 is used to organize the content, CSS3 is used for visual design and responsive layouts, and JavaScript is used for interactive behaviour. Git and GitHub are used for source-code management, while GitHub Pages is used to publish the website online.

The website includes Home, About, Education, Skills, Projects, Experience, CV and Contact areas. It also provides direct access to my GitHub and LinkedIn profiles. This report explains how System Analysis and Design concepts were used to identify the problem, define requirements, model user interaction, plan the interface, implement the solution and evaluate the final website.

1. Introduction & Problem Statement

1.1 Project Context

As an ICT undergraduate, my academic information, technical skills, projects, CV and professional profiles can easily become spread across several different places. For example, source code may be stored on GitHub, professional information may be available on LinkedIn, and academic details may be included in a separate CV. This makes it less convenient for a lecturer, recruiter or collaborator to understand my overall profile quickly.

I developed this portfolio website to bring the most important information together in one place. It also gives me a practical way to demonstrate the front-end development knowledge that I have gained while studying ICT.

1.2 Problem Statement

A traditional CV can present qualifications and skills, but it provides limited space to demonstrate practical project work and technical ability. Visitors may also need to open several separate links to view source code, professional profiles and project information. Therefore, the main problem was the lack of a single, responsive and maintainable online professional profile that could organize my education, skills, projects, CV and contact information clearly.

1.3 Proposed Solution

The proposed solution is a responsive professional portfolio website with section-based navigation. Visitors can view my background, education, skills, projects and experience, access my CV, and open my GitHub and LinkedIn profiles. The system is designed as a lightweight front-end solution using HTML5, CSS3 and JavaScript, without requiring a database or complex backend. The website is maintained through GitHub and deployed using GitHub Pages.

2. Objectives, Stakeholders and Scope

2.1 Objectives

- Create a professional online portfolio that represents me as an ICT undergraduate.

- Present my education, technical skills, soft skills, projects and experience in an organized format.
- Provide easy access to my CV, GitHub profile and LinkedIn profile.
- Demonstrate practical knowledge of HTML5, CSS3 and JavaScript.
- Create a responsive interface that works on desktop, tablet and mobile screen sizes.
- Apply System Analysis and Design concepts to a real web-development project.
- Keep the website simple to maintain and update as I complete new projects and develop new skills.

2.2 Stakeholders

Stakeholder	Role / Interest
Primary Users	Recruiters, employers, technical reviewers and potential collaborators who may evaluate my skills and project work.
Secondary Users	Lecturers, academic supervisors and fellow students who may review the portfolio and SAD assignment.
System Owner	K.M.N.M.R.S.R. Thilakarathna, responsible for maintaining personal information, projects, CV and professional links.

2.3 Scope

In scope: Home, About, Education, Skills, Projects, Experience, CV and Contact content; responsive navigation; professional external links; project presentation; CV access; and responsive front-end design.

Out of scope: user authentication, login accounts, a database, a content management system, complex server-side processing and backend analytics.

2.4 Constraints and Assumptions

The project was developed within the limited time available for the SAD assignment and uses mainly free front-end technologies and GitHub services. The current website is designed so that its content can be expanded later with additional projects, skills and professional experience.

3. Requirements Analysis

3.1 Functional Requirements

ID	Requirement	Priority
FR-01	Display my name, professional title and introduction.	High
FR-02	Allow visitors to navigate between the main portfolio sections.	High
FR-03	Provide responsive navigation for desktop and mobile devices.	High
FR-04	Display education, technical skills and soft skills.	High
FR-05	Present project information and relevant project links.	High
FR-06	Provide access to my professional CV.	High
FR-07	Provide links to my GitHub and LinkedIn profiles.	High
FR-08	Display contact information for	Medium

	visitors.	
--	-----------	--

3.2 Non-Functional Requirements

Attribute	Requirement
Usability	The interface should be easy to understand, readable and simple to navigate.
Responsiveness	The layout should adapt to desktop, tablet and mobile screen sizes.
Performance	The site should remain lightweight by using standard front-end technologies.
Compatibility	The portfolio should work on commonly used modern web browsers.
Maintainability	Personal information, projects and skills should be easy to update.
Consistency	Typography, spacing, buttons and content cards should follow a consistent visual style.

4. System Modeling

4.1 Use-Case Diagram

The primary actor of the portfolio is the Visitor. A visitor may be a recruiter, lecturer, fellow student, technical reviewer or collaborator.

Use Case	Description
UC-01 - Navigate Portfolio	Visitor uses the navigation menu or scrolling to move between portfolio sections.
UC-02 - Review Professional Profile	Visitor reads About, Education, Skills and Experience information.
UC-03 - Review Projects	Visitor explores project information and available project links.
UC-04 - Access CV	Visitor accesses the professional CV from the CV area.
UC-05 - Open External Profiles	Visitor opens GitHub or LinkedIn for additional professional information.
UC-06 - Access Contact Information	Visitor views contact details when they want to communicate with me.

[Insert Use-Case Diagram Here]

4.2 Activity Diagram

The main activity flow is: Start -> Open Portfolio -> View Home -> Select a Section -> Review Profile / Education / Skills / Projects -> Access CV or External Profiles -> View Contact Information -> Leave Website -> End.

[Insert Activity Diagram Here]

5. System Design

5.1 Site Map / Information Architecture

- Home - introduction, professional identity and main actions.
- About - personal introduction and career information.
- Education - degree and academic background.

- Skills - technical skills, tools and soft skills.
- Projects - project descriptions, technologies and available links.
- Experience - academic or practical experience.
- CV - access to the professional CV.
- Contact - contact details and professional profile links.

5.2 Interface Design Decisions

The portfolio uses a modern technology-focused interface suitable for an ICT undergraduate. A clear visual hierarchy is used so visitors can quickly identify headings, descriptions, skills, projects and important actions. Structured sections and card-based content help separate information and keep the interface readable.

The navigation remains easy to access, while the layout adapts for smaller screens. Buttons and links are used for important actions such as opening project information, accessing the CV and visiting professional profiles. Consistent spacing, typography and interactive elements help maintain a professional appearance throughout the website.

5.3 Wireframes

Wireframes were used to plan the basic structure and placement of important interface elements before the final implementation. The wireframe should represent the Home, About, Education, Skills, Projects, Experience, CV and Contact sections.

[Insert Figma Wireframe Screenshot Here]

6. Implementation Details

The portfolio was implemented as a front-end website using HTML5, CSS3 and JavaScript. Git and GitHub are used for version control and repository management, while GitHub Pages is used to publish the live website.

Technology	Implementation
HTML5	Defines the semantic structure, navigation and main portfolio sections.
CSS3	Controls typography, spacing, responsive layouts, cards, buttons and visual presentation.
JavaScript	Supports interactive user-interface behaviour.
Git & GitHub	Used for source-code version control and repository management.
GitHub Pages	Used to deploy the portfolio as a publicly accessible website.

6.1 Key UI/UX Features

- Responsive navigation for desktop and smaller screens.
- Clear hero section containing my professional identity.
- Structured About and Education information.
- Skills presented in organized categories.
- Project presentation with relevant information and links.
- Dedicated CV access area.
- GitHub, LinkedIn and contact links.
- Consistent responsive design across the website.

7. Testing, Quality Assurance and Deployment Evaluation

ID	Test	Expected Result	Review Result	Status
TC-01	Section Navigation	Navigation links move to the correct sections.	Main navigation corresponds with portfolio sections.	Pass
TC-02	Mobile Navigation	Navigation remains usable on smaller screens.	Responsive navigation is provided.	Pass
TC-03	Responsive Layout	Content adapts to different screen sizes.	Layout reorganizes according to available space.	Pass
TC-04	Project Links	Project controls open intended destinations.	Available project links can be accessed.	Pass
TC-05	CV Access	CV control provides access to the CV.	CV area provides access to professional CV.	Pass
TC-06	External Profiles	GitHub and LinkedIn links open correct destinations.	Professional links are included.	Pass
TC-07	Live Deployment	Portfolio is publicly accessible online.	Portfolio is deployed using GitHub Pages.	Pass

8. Reflection and Conclusion

8.1 SAD Learning Reflection

Developing this portfolio helped me understand that even a small front-end system benefits from a structured System Analysis and Design process. Identifying the problem and requirements before implementation made it easier to decide what information the website should contain and how visitors should interact with it.

Use cases and activity modelling helped me think about the visitor journey, while the site map and wireframe helped organize the content before implementation. Testing also showed that a professional portfolio is not only about appearance; navigation, responsiveness, links and access to important information must work correctly.

8.2 Career Value

The portfolio provides a professional platform that can grow together with my academic and technical experience. Instead of sharing only a traditional CV, I can use one website to present my education, skills, projects, experience, CV and professional profiles. I can continue updating it as I complete more projects and prepare for internships and future entry-level opportunities.

8.3 Conclusion

The Professional Portfolio Website provides a responsive online platform for presenting my academic background, technical and soft skills, projects, experience, CV and contact information. The project uses HTML5, CSS3 and JavaScript, is maintained through Git and GitHub, and is deployed using GitHub Pages.

Through this project, I was able to apply System Analysis and Design concepts to a practical web-development task while creating a portfolio that can support my future academic and professional development.

Portfolio Evidence

Live Website: <https://shenalthilakarathana2003-eng.github.io/SAD-Portfolio/>

GitHub Repository: <https://github.com/shenalthilakarathana2003-eng/SAD-Portfolio>

LinkedIn Profile: <https://www.linkedin.com/in/shenal-thilakarathne-9539a7343>